

Tutorial 10

Indian Institute of Technology Jodhpur
Mathematics-I (MAL1010)

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1. Using the definition of partial derivatives, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for the function $f(x, y) = 2x^2 - xy + 2y^2$ at $(1, 2)$.
2. Using the definition of partial derivatives, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(0, 0)$ for the function

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}.$$

3. Using the definition of partial derivatives, find $\frac{\partial^2 f}{\partial y \partial x}$ and $\frac{\partial^2 f}{\partial x \partial y}$ at $(0, 0)$ for the function

$$f(x, y) = \begin{cases} \frac{xy^5}{x^2+y^4}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}.$$

4. Check whether the following function is differentiable or not at $(0, 0)$.

$$f(x, y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}.$$

5. Prove that $f(x, y) = x^2 + y^2$ is differentiable at every point of $\mathbb{R}^2$.
6. Prove that $f(x, y) = \cos(x + y)$ is differentiable at $(\frac{\pi}{4}, \frac{\pi}{4})$.
7. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = \begin{cases} \frac{x^2y}{x^2+y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}.$$

Show that the directional derivative $\mathbf{D}_u f(0, 0)$ exists for any unit vector u .

8. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = \begin{cases} \frac{x^3-y^3}{x^2+y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}.$$

Show that the directional derivative $\mathbf{D}_u f(0, 0)$ exists for any unit vector u .